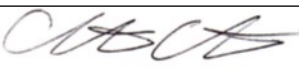


**Guideline Title: Pediatric Seizures/Status Epilepticus**

**Guideline Number: GUID-3203-PED015**

|                                                                                              |                             |
|----------------------------------------------------------------------------------------------|-----------------------------|
| Issue date: 09/09/2014                                                                       | Replaces Dept. Policy:      |
| Revision dates: 4/30/24                                                                      | Developed by: Air Care Team |
| Approved by: Dr. Christopher Hunter, MD<br>Air Care Team Medical Director                    | Approved by:                |
| Signature:  | Signature:                  |
| Department Numbers: 3203                                                                     |                             |

- I. **PURPOSE:** Emergency management is imperative for status epilepticus. The goal is to stop the seizure, identify and correct the underlying cause, and to maintain a patent airway while providing oxygenation and protecting the patient from injury.
- II. **PATHOPHYSIOLOGY:** Status epilepticus is a single generalized tonic-clonic seizure or series of convulsions lasting 30 or more minutes during which time the patient does not regain consciousness. It may also be associated with permanent neuronal destruction.

Causes of acute seizures include Toxic/metabolic disturbances (hypernatremia, hyponatremia, hypocalcemia, hypomagnesemia, hypoglycemia, exogenous toxins, and inherited metabolic disorders), intracranial pathology (hemorrhage, arteriovenous malformations, aneurysms, brain tumor), anoxia, hypertension, trauma, withdrawal from drugs/alcohol, or change/withdrawal of medications in children with known seizures.

Causes of chronic seizures include Idiopathic, post-anoxia injury, post-infectious injury, post-traumatic injury, post-hemorrhagic injury, post-toxic injury, degenerative diseases, or congenital disorders.

Classification of seizures

- Partial (focal) seizures : Simple partial seizures or Complex partial seizures.
- Generalized seizures: convulsive or non-convulsive, tonic, clonic, atonic, myoclonic.
- Unclassified epileptic seizures: all seizures that cannot be classified.

III. **DEFINITIONS:**

IV. **DEPARTMENT GUIDELINE:**

- A. Prepare a safe environment for the patient. Protect patient from injury.
- B. Assess and manage airway, breathing, and circulation as per *GUID3203-PED001 – Pediatric General Management*. Optimize oxygen content.
- C. If a patient requires airway management per *GUID3203-G008 Rapid Sequence Intubation (RSI) for Endotracheal (ET) Intubation*, avoid long-acting paralytics which may mask ongoing seizure activity.
- D. Establish IV/IO access.
- E. If the blood sugar level is less than 60 d/L, treat per *GUID3203-M021 – Hypoglycemia*.
- F. If the patient is actively seizing, administer one of the following drugs
  - i. **Midazolam 0.05-0.2 mg/kg IV/IM/IO/IN** repeat as necessary q 2-5 minutes - Max single dose 5mg for IV/IN and 10mg of IM). No max total dose.

**Guideline Title: Pediatric Seizures/Status Epilepticus**

**Guideline Number: GUID-3203-PED015**

- ii. **Lorazepam at 0.05-0.1 mg/kg IV/IO**, Max dose 4mg and may repeat dose in 5 minutes. Repeat as needed – No max total dose.
  - iii. **Diazepam 0.5 mg/kg PR**. May repeat q 10-15 min.
- G. If seizure activity continues, administer **Levetiracetam 60 mg/kg IV/IO** Maximum dose 4.5 grams over 5 minutes. For doses >1000 mg, dilute vials in 100 ml of 0.9% NS and administer over 15 minutes. *This is the preferred second line agent after benzodiazepines.*
- H. If the patient has required multiple doses of the above medications, consult PICU MD to obtain an order for administration of:
  - i. **Phenobarbital 15-20 mg/kg IV** (not faster than 30 mg/min). May repeat every 15 min as needed. Max total dose 1000 mg.
    - a. Because the risk of respiratory failure is high with the completion of the phenobarbital load, the infusion should be interrupted if the seizure stops before its completion.
  - ii. **Fosphenytoin bolus loading dose of 10-20 mg PE/kg IV infusion at 50 mg PE/minute** (max dose 1500 mg).
  - iii. **Valproic acid 20-40 mg/kg IV/IO** (max dose 3000mg)
- I. Reassess airway patency.
- J. If at sending facility and febrile seizure is suspected, determine last dose of antipyretic, and consider dose of rectal **Acetaminophen 20 mg/kg PR** every 4 to 8 hours.
- V. **DOCUMENTATION:**
  - A. EMS Charts
- VI. **REFERENCES:**
  - A. Pediatric Advanced Life Support Manual, 2020. American Heart Association.